Sheet 11 Mon. 16 Nov. 2026 Collinearity, parallelism, intersection of lines

1. Warm-up · Flash Is point $(3;5)$ lie on the line $y = 2x - 1$?

2. Core · Method Relative positions of $d : y = 3x + 1$ with $d' : y = 3x - 2$, then with $d'' : y = -x + 1$.

3. Core + · Method Show that $A(1;4)$, $B(2;7)$ and $C(4;13)$ are collinear in two ways.

4. Review · Flash On a map with scale $\frac{1}{50\,000}$, two towns are 4 cm apart. What is the actual distance?

5. Running thread · Method Procedure: choose a number x ; add 3; multiply by 2; subtract 6. Calculate the result for $x = 5$, then write an expression in x and simplify it. What do you notice?

Sheet 12 Tue. 17 Nov. 2026 Intersection of two lines; Python collinearity function

1. Warm-up · Flash Solve 2 $x + 1 = -x + 7$.

2. Core · Method Coordinates of the intersection of $y = 2x + 1$ and $y = -x + 7$.

3. Core + · Reading code Outputs of alignes(0,0,1,2,2,4) and alignes(0,0,1,2,2,5) ?

```
def alignes(xa, ya, xb, yb, xc, yc):
    return (xb-xa)*(yc-ya) - (yb-ya)*(xc-xa) == 0
```

4. Review · Expand Expand and simplify 3 $(x - 2)$.

5. Running thread · Prove Show that the sum of three consecutive integers is a multiple of 3.

Real numbers and intervals

Sheet 13 Thu. 19 Nov. 2026 Sets of numbers: $\mathbb{N}, \mathbb{Z}, \mathbb{D}, \mathbb{Q}, \mathbb{R}$

1. Warm-up · Flash Give the decimal expansion of $\frac{3}{8}$ and an approximate value of $5\frac{}{6}$ to 10^{-3} accuracy.

2. Core · Method Give the smallest set ($\mathbb{N}, \mathbb{Z}, \mathbb{D}, \mathbb{Q}, \mathbb{R}$) containing each of these numbers: -4 ; $0{,}25$; $\frac{1}{3}$; $\sqrt{2}$; $\frac{6}{3}$.

3. Core + · True/false True or false? Justify. The number $\frac{1}{3}$ is a decimal number.

4. Review · Flash Calculate $(x + 1)(x - 3)$ for $x = 2$.

5. Running thread · Flash How many faces, vertices and edges does a right triangular prism have?

Sheet 14 Mon. 23 Nov. 2026 Decimal bounds and rounding

1. Warm-up · Flash Round 3.14159 to 10^{-2} accuracy, then to 10^{-3} accuracy.

2. Core · Method Bound $\overline{2}$ between two decimals differing by 10^{-2}.

3. Core + · Reading code What is the final value of `k` ? What can be deduced about $\sqrt{2}$?

```
k = 10
while (k/10)**2 < 2:
    k = k + 1
```

4. Review · Flash 5 kg of rice cost 12,500 Ar. How much is 8 kg?

5. Running thread · True/false True or false? Justify. A quadrilateral whose diagonals have the same length is a rectangle.